



INTRODUCTION TO PRODUCTION AND MANUFACTURING SYSTEMS

INDU 211

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Agenda

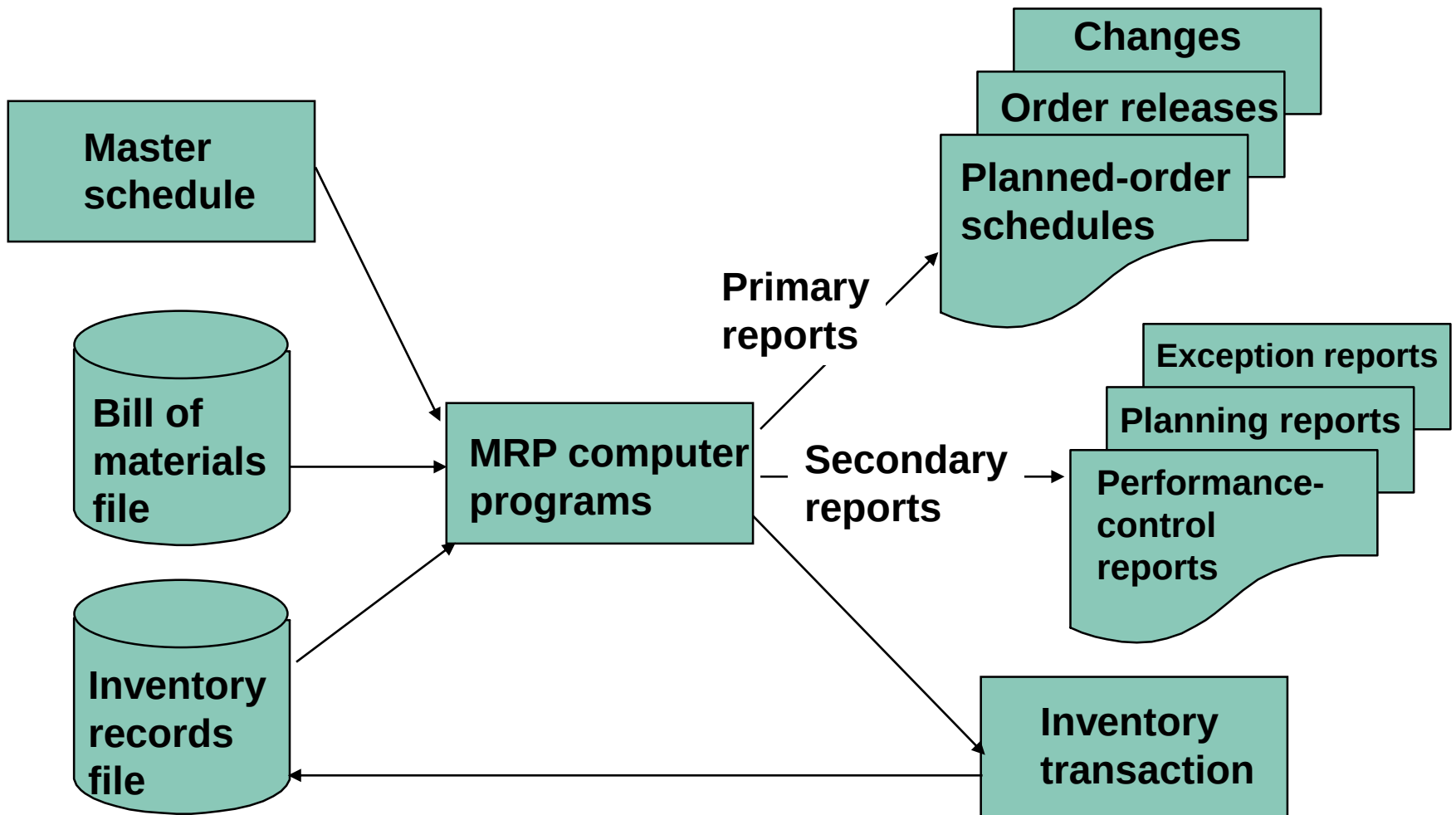
- Chapter 7 Operations Planning and Control

Materials Requirement Planning - MRP

- A technique used by order planners to determine parts needed to meet a known master schedule for end items
- Computer based production and inventory control system that attempts to minimize inventories while maintaining adequate materials for the production process

Materials Requirement Planning - MRP

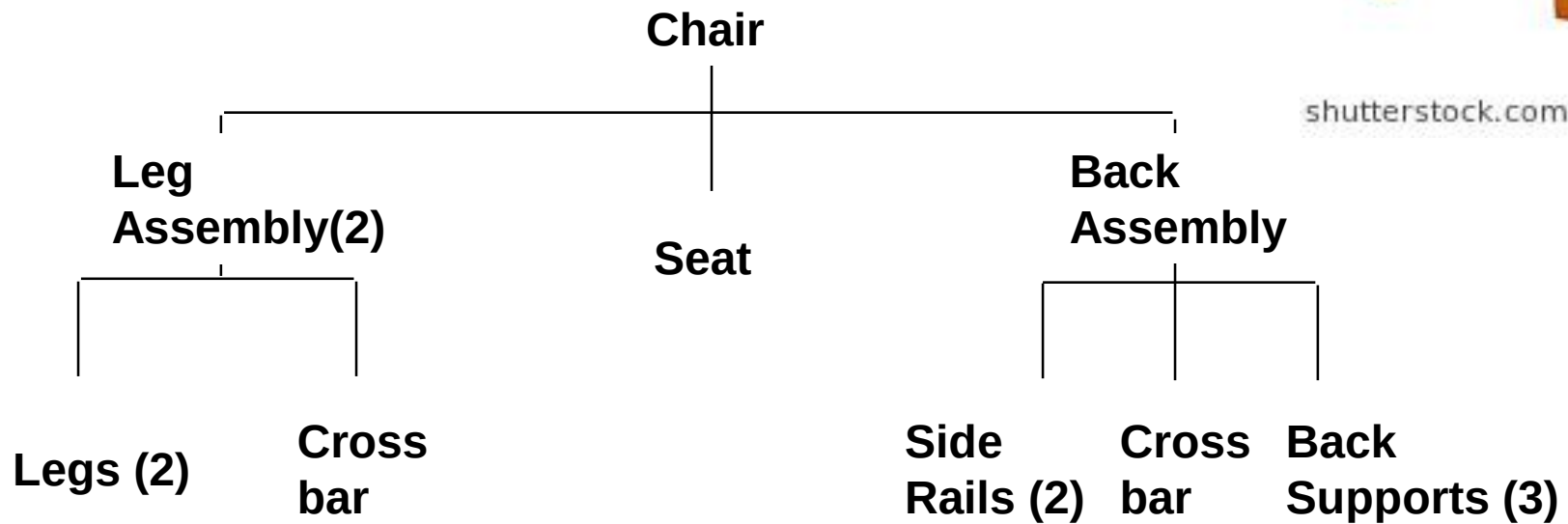
MRP Inputs



Product Structure Tree



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Master Schedule		Week Number	1	2	3	4	5
		Quantity					
ITEM		Gross requirements					
		Scheduled receipts					
Quantity =		Projected on hand					
LT =		Net requirements					
Lot Size =		Planned-order receipts					
		Planned-order releases					
ITEM		Gross requirements					
		Scheduled receipts					
Quantity =		Projected on hand					
LT =		Net requirements					
Lot Size =		Planned-order receipts					
		Planned-order releases					
ITEM		Gross requirements					
		Scheduled receipts					
Quantity =		Projected on hand					
LT =		Net requirements					
Lot Size =		Planned-order receipts					
		Planned-order releases					

Example 1

P1 200 (week 17)

SA1	400
SA2	200
C1	1,600
C2	800
C3	400
C4	200
C5	3,200

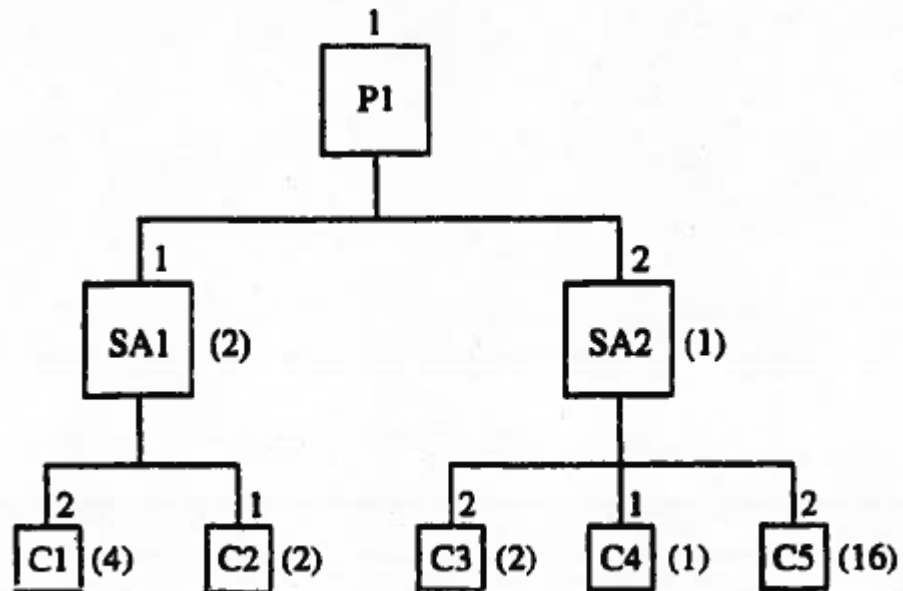


Fig. 7.10. Product structure diagram for product P1.

C4 75 units (scheduled to arrive in week 12)

C5 716 (inventory on hand- beginning of week 10)

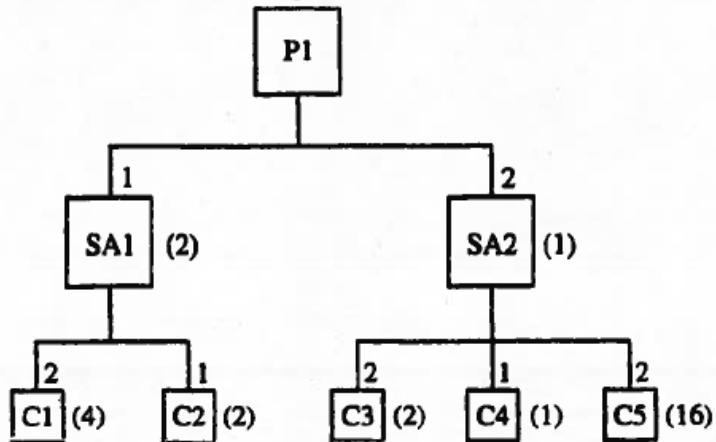


Fig. 7.10. Product structure diagram for product P1.

P1 200 (week 17)

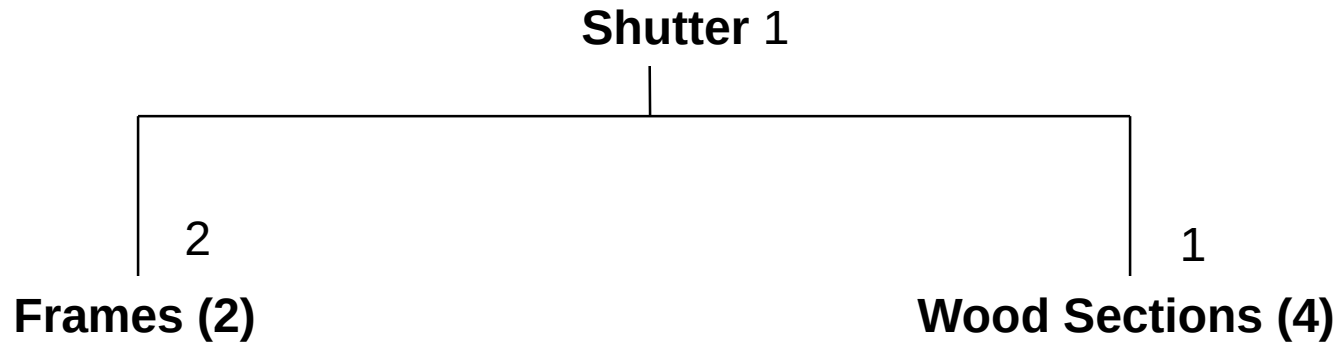
SA1 400
SA2 200
C1 1,600
C2 800
C3 400
C4 200
C5 3,200

C4 75 units (scheduled to arrive in week 12)

C5 716 (inventory on hand- beginning of week 10)

Period	10	11	12	13	14	15	16	17	18	19
Item: P1 LT 1										
Gross Requirements								200		
Scheduled Receipts										
On Hand	0									
Net Requirements								200		
Planned Order Releases							200			
Item: SA 2 LT 2										
Gross Requirements							200			
Scheduled Receipts										
On Hand	0									
Net Requirements							200			
Planned Order Releases					200					
Item: C3 LT 2										
Gross Requirements					400					
Scheduled Receipts										
On Hand	0									
Net Requirements					400					
Planned Order Releases			400							
Item: C4 LT 1										
Gross Requirements					200					
Scheduled Receipts			75							
On Hand	0		75	75	75					
Net Requirements					125					
Planned Order Releases			125							
Item: C5 LT 2										
Gross Requirements					3200					
Scheduled Receipts										
On Hand	716	716	716	716	716	716				
Net Requirements					2484					
Planned Order Releases			2484							
Item: LT										
Gross Requirements										
Scheduled Receipts										
On Hand										
Net Requirements										
Planned Order Release										

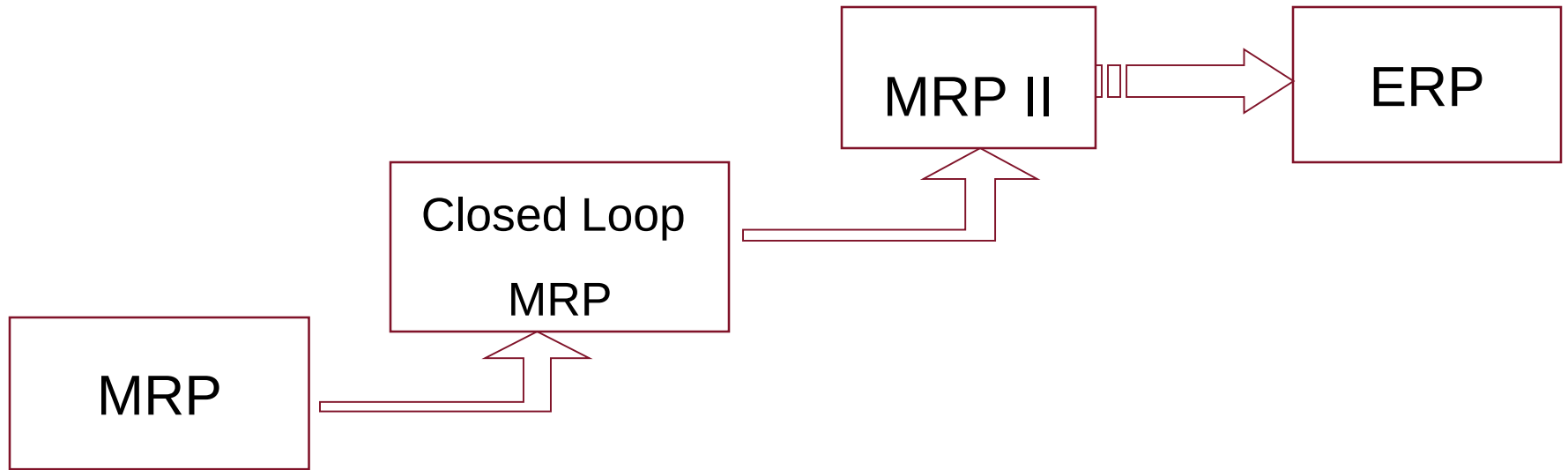
Example 2



- Develop an MRP plan for shutters, frames and wood sections
- 100 shutters are needed at the start of week 4 and 150 are needed at the start of week 8. Shutters are ordered with lead time (LT) of 1 week.
- Frames are ordered and the lead time (LT) is 2 weeks. Frames are ordered in lots of 320 units.
- Wood sections are made by the firm and fabrication takes 1 week, Wood sections are made in lots of 70.
- A scheduled receipt of 70 wood sections at the beginning of Week 1.

Master Schedule		Week Number	1	2	3	4	5	6	7	8
for Shutters:		Quantity	100150							
Shutters:		Gross requirements	0	0	0	100	0	0	0	150
		Scheduled receipts								
Quantity =	1	Projected on hand	0	0	0	0	0	0	0	0
LT =	1	Net requirements	0	0	0	100	0	0	0	150
Lot Size =	1	Planned-order receipts	0	0	0	100	0	0	0	150
		Planned-order releases	0	0	100	0	0	0	150	0
Frames:		Gross requirements	0	0	200	0	0	0	300	0
		Scheduled receipts								
Quantity =	2	Projected on hand	0	0	0	120	120	120	120	140
LT =	2	Net requirements	0	0	200	0	0	0	180	0
Lot Size =	320	Planned-order receipts	0	0	320	0	0	0	320	0
		Planned-order releases	320	0	0	0	320	0	0	0
Wood sections:		Gross requirements	0	0	400	0	0	0	600	0
		Scheduled receipts	70							
Quantity =	4	Projected on hand	70	70	70	20	20	20	20	50
LT =	1	Net requirements	0	0	330	0	0	0	580	0
Lot Size =	70	Planned-order receipts	0	0	350	0	0	0	630	0
		Planned-order releases	0	350	0	0	0	630	0	0
Note: Use Lot Size = 1 for lot-for-lot ordering.										

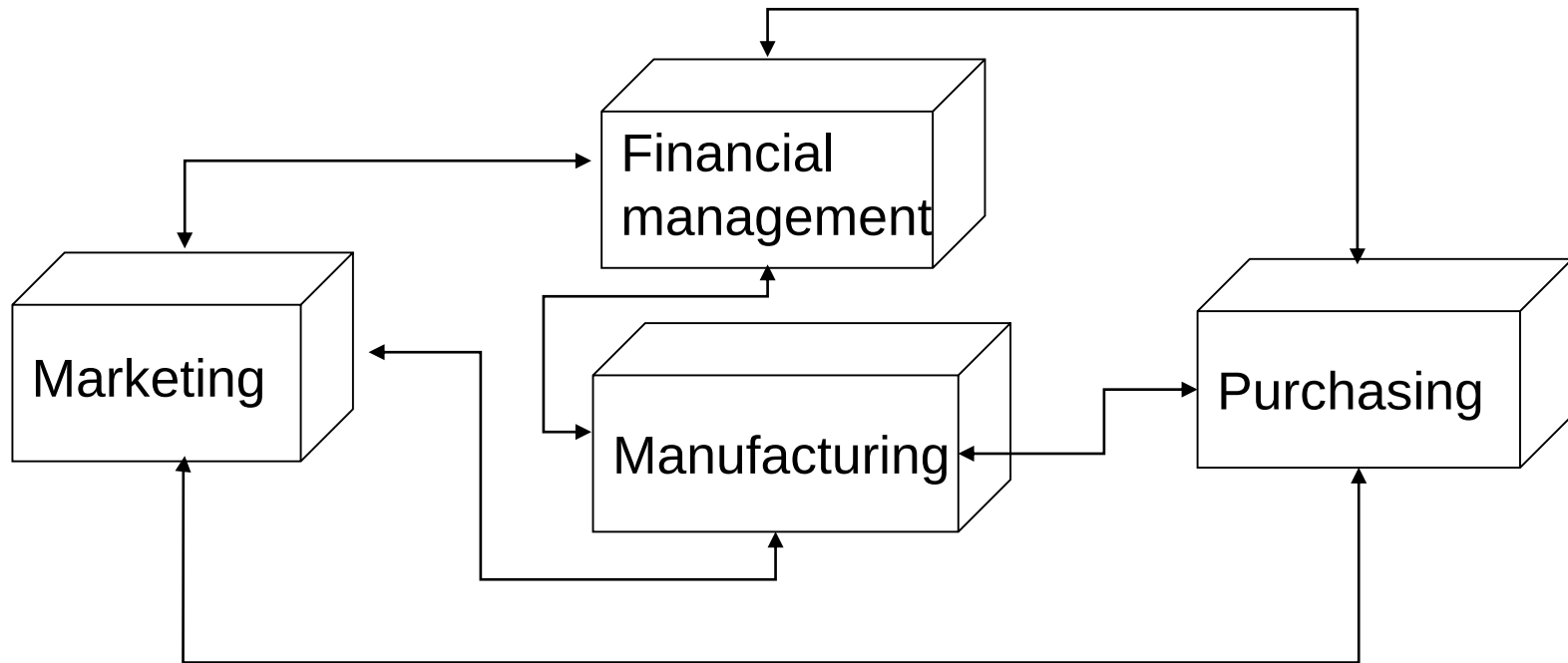
Evolution of Planning Systems



Closed-loop MRP

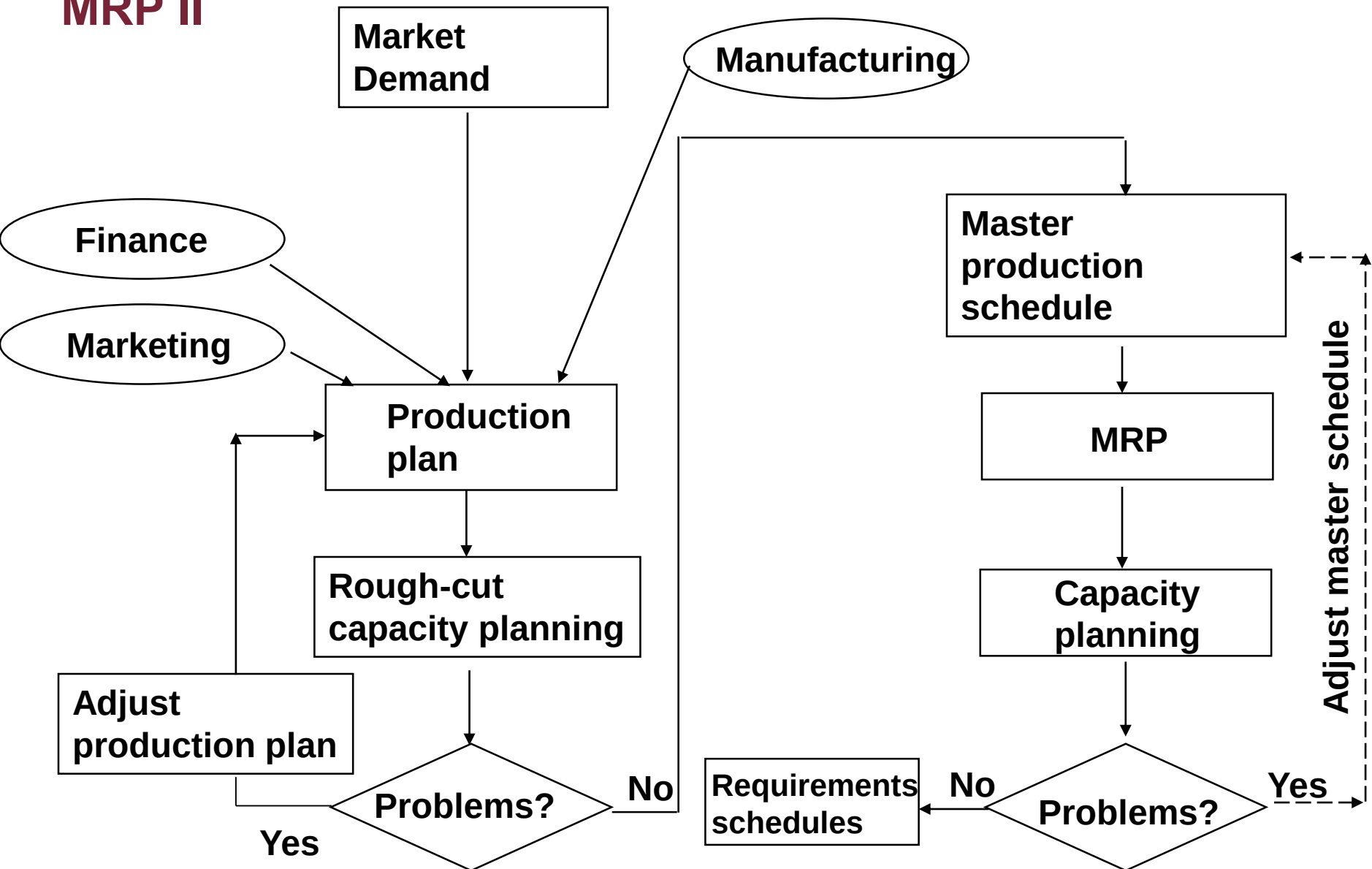
- A system used for production planning and inventory control, with an information feedback feature
 - Enables plans to be checked and adjusted
- Synchronize the purchasing or materials procurement plans with the master production schedule
- Software application
- The system feeds back information about completed manufacture and materials on hand into the MRP system
 - Plans can be adjusted according to capacity and other requirements

Manufacturing Resource Planning (MRP II)



Expanded MRP: emphasis placed on integration
Simulation component (to answer what-if questions)

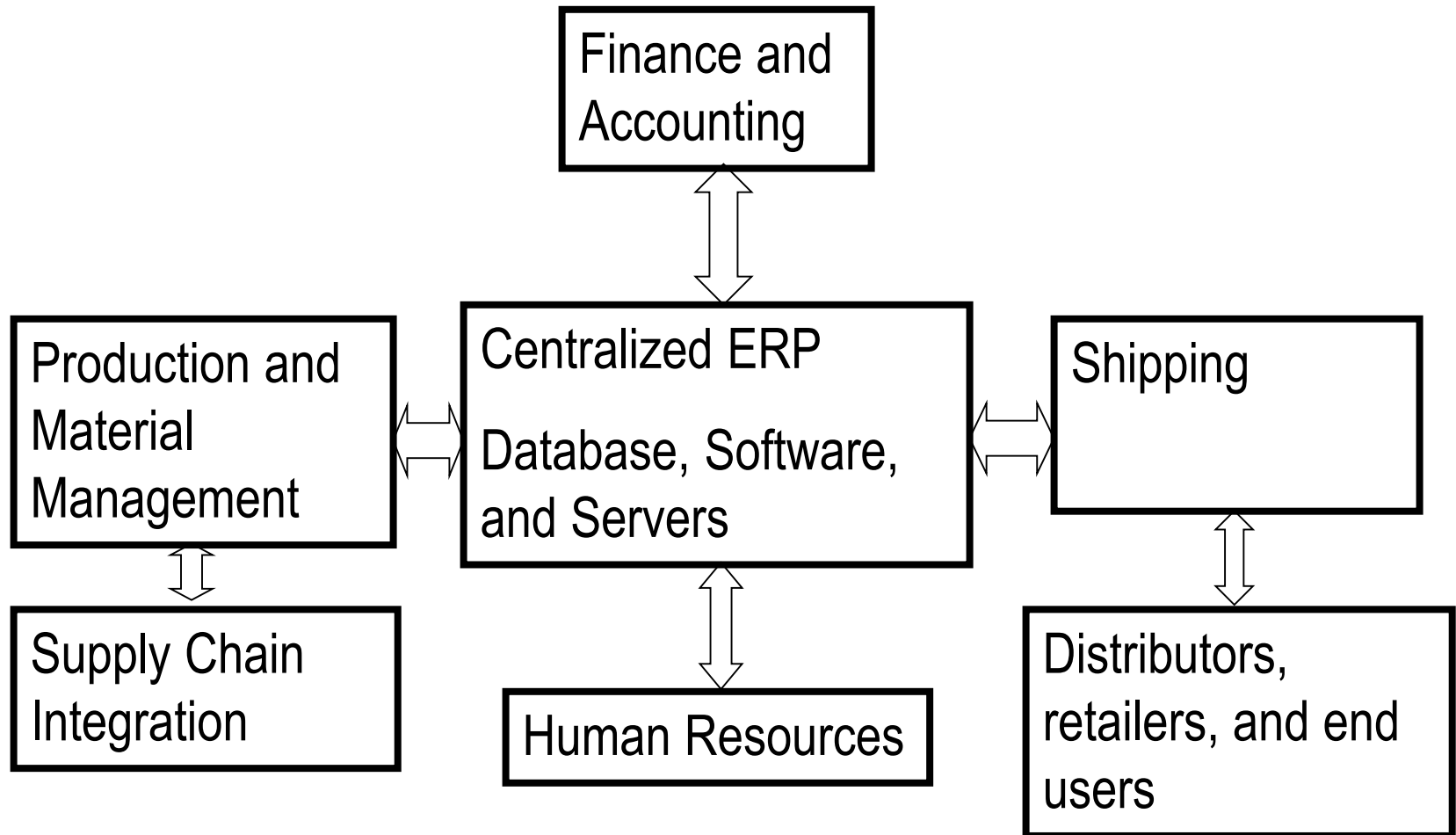
MRP II



ERP

- Enterprise Resource Planning (ERP)
 - An MRP II system that ties customer orders to enterprise-wide resources and suppliers
 - Integrates enterprise activities completely
 - Product planning, development, manufacturing, sales and marketing integrated in a single database, application and user interface
 - Information system that facilitates information sharing and decision-making across the enterprise

Example of an ERP system



ERP

- Before ERP
 - Many different systems
 - Paper and human integration
 - Varying levels of technology infrastructure
 - Varying levels of user competence
- Benefits of ERP
 - Integrated systems for clients in multiple locations
 - Reduced inventory
 - Timely responsiveness
 - Faster delivery lead times
- Drawbacks of ERP
 - Complex
 - Expensive
 - Improper implementation, very long
 - Technology-dependent
 - More responsibility, accountability

ERP Statistics for Typical Implementation

- Average implementation time: **1 to 3 years**
- Average Total Cost of Ownership (including hardware, software, professional services, internal staff costs) was **\$15 million**
- Payback: **8 months (31 months total)**
- Annual savings is **\$1.6 million/year**

Operations scheduling

- Assign specific operations to specific operating facilities with specific start and end times
- Minimized cost
 - WIP inventory
 - Idle operators
 - Idle equipment
 - Over-time
 - Lateness
- Scarce resources (machines)
- External random perturbations
 - Canceled/emergency order
 - Late delivery of material
 - Machine/tool breakdown

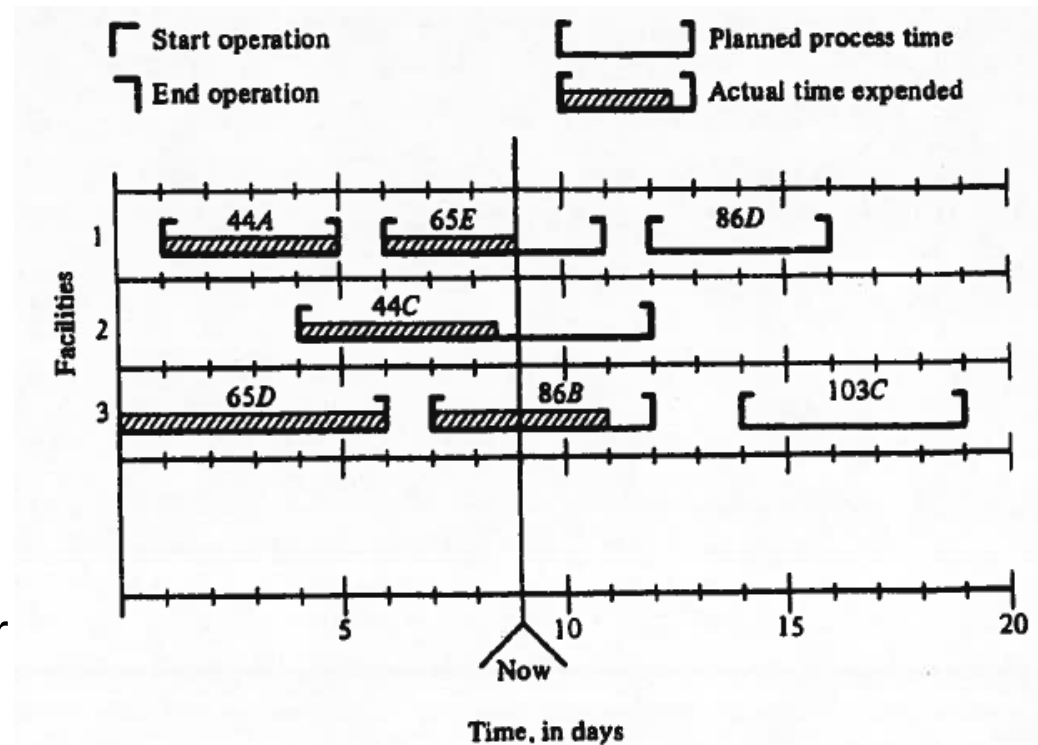


Fig. 7.9. Gantt scheduling chart.

Dispatching and progress control

- How all decisions made on production and inventory plan and operations schedules are implemented (transferred into action)?
- Relaying plans (production control office) to shop floor supervisors
- Recording progress
- Analyzing deviations from the plan
- Corrective actions
- Information systems

Lean manufacturing

- A systematic method for the elimination of waste ("Muda") within a manufacturing system
- Waste is ‘anything other than the minimum amount of equipment, materials, parts, space, and worker’s time, which are absolutely essential to add value to the product.’

— Shoichiro Toyoda
President, Toyota

Types of Waste

- Overproduction
- Waiting
- Transportation
- Inefficient processing (over processing due to poor tool or product design)
- Inventory
- Unnecessary motion
- Product defects

What is Just-in-Time?

- Management philosophy of continuous and forced problem solving
- Supplies and components are ‘pulled’ through system to arrive where they are needed when they are needed.

Push versus Pull

- Push system: material is pushed into downstream workstations regardless of whether resources are available
- Pull system: material is pulled to a workstation just as it is needed

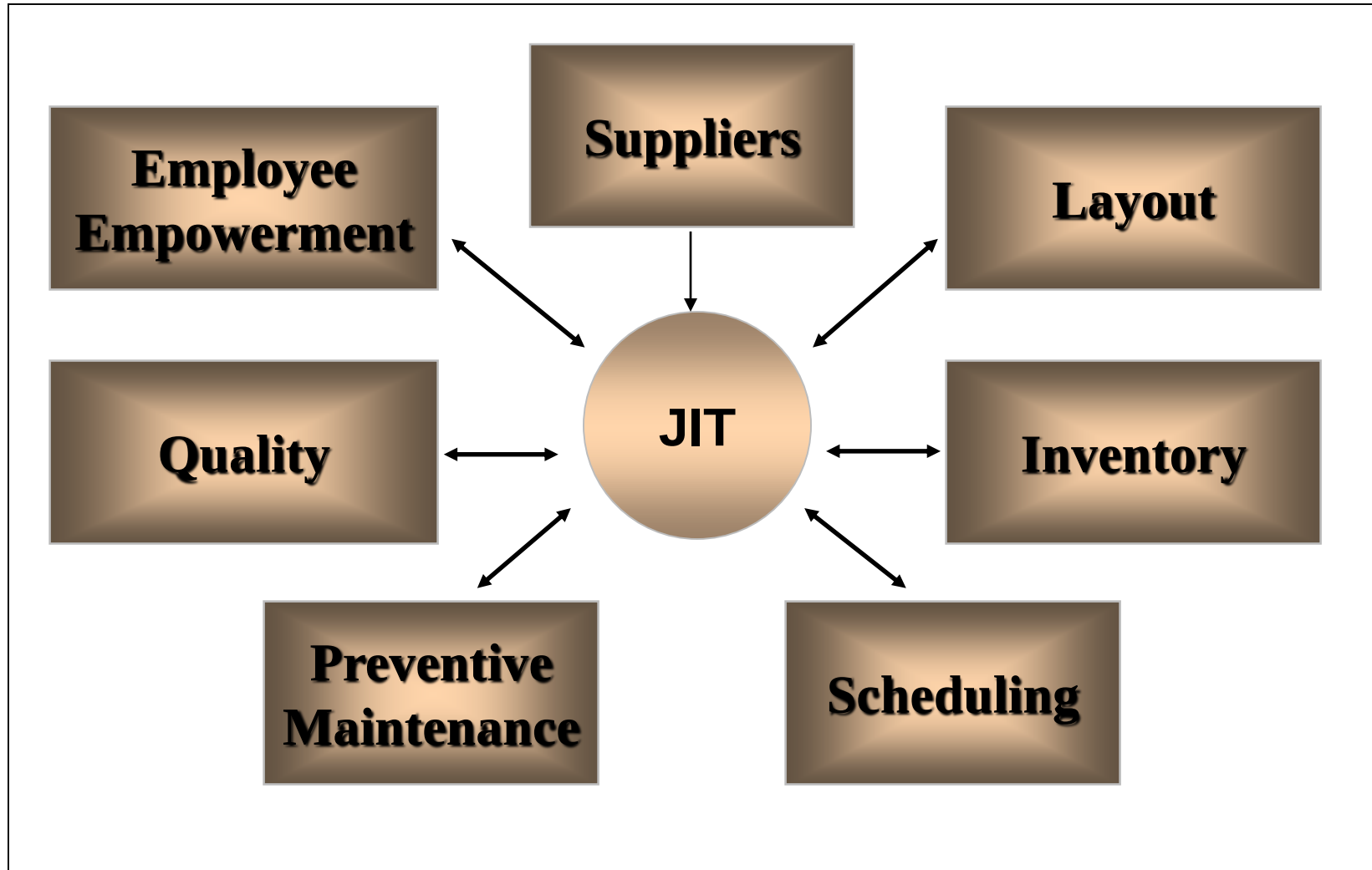
What Does Just-in-Time Do?

- Attacks waste
 - Anything not adding value to the product
 - Customer's perspective
- Exposes problems and bottlenecks caused by variability
 - Deviation from optimum
- Achieves streamlined production
 - By reducing inventory

Variability Occurs Because

- Employees, machines, and suppliers produce units that do not conform to standards, are late, or are not the proper quantity
- Engineering drawings or specifications are inaccurate
- Production personnel try to produce before drawings or specifications are complete
- Customer demands are unknown or uncertain

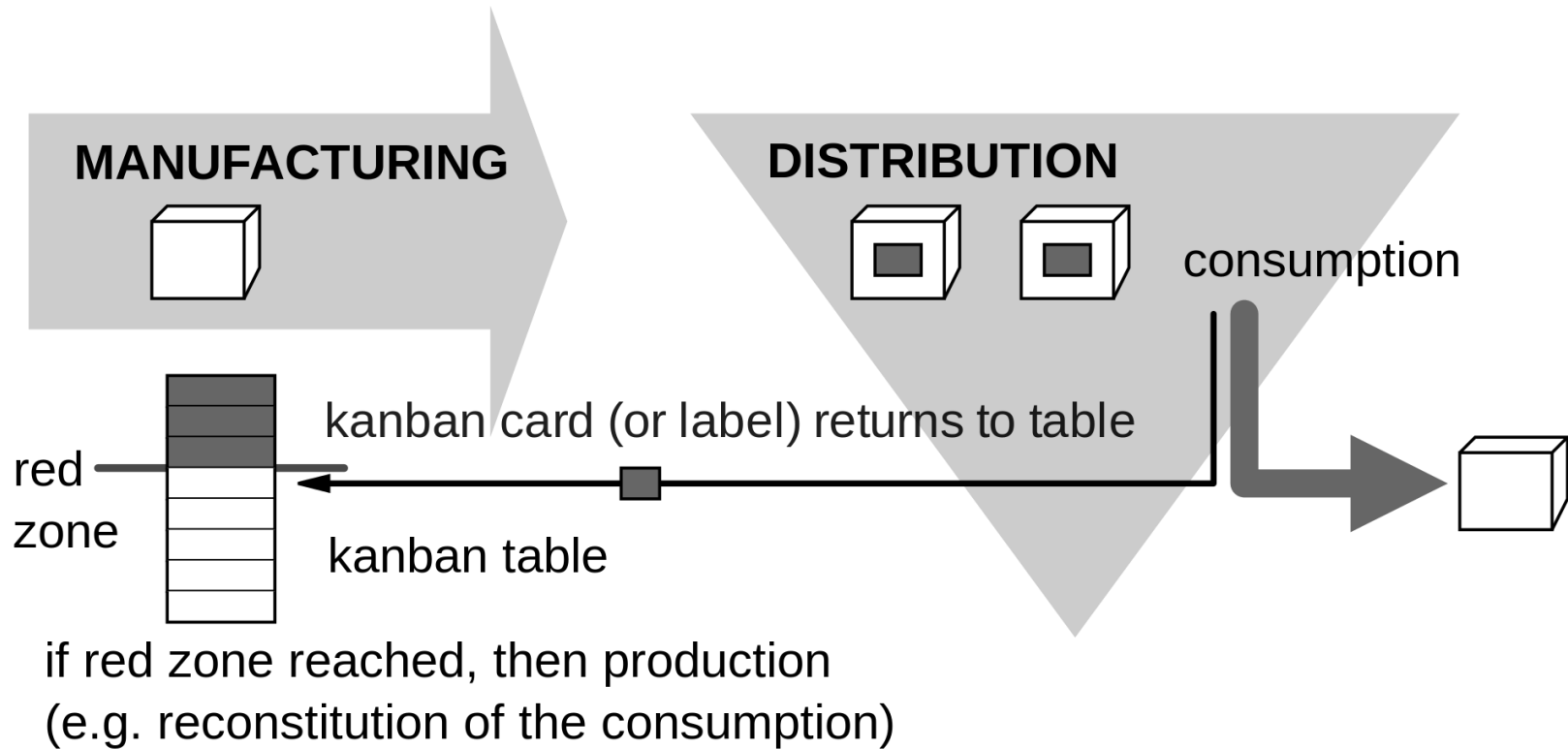
Just-in-Time Success Factors



Kanban

- Japanese word for card
- a scheduling system for lean manufacturing and just-in-time manufacturing (JIT)
- **Kanban** is an inventory-control system to control the supply chain
- A signal is sent to produce and deliver a new shipment as material is consumed
 - uses the rate of demand to control the rate of production
- These signals are tracked through the replenishment cycle and bring visibility to suppliers and buyers
- Authorizes production from downstream operations
- May be a card, flag, verbal signal, etc.
- Used often with fixed-size containers
 - Add or remove containers to change production rate

Kanban system



To achieve JIT

- Smooth material flow
- Reduction of setup times
- Reduction of vendor lead times
- Zero-defect components
- Disciplined shop floor control

Comparing MRP and Kanban?

- Results of MRP, if implemented as computed, should be “optimal”
- Results based on a Kanban system are not aiming at “optimal”
- In a not-so-accurate comparison, MRP is a top-down approach while Kanban is a bottom-up approach